

TM3 Advanced SQL Mini Project - Solutions

1. Create EMP_DETAILS

```
CREATE TABLE LODGE_DETAILS(  
    LODGE_NAME VARCHAR2(30) PRIMARY KEY,  
    LODGE_MANAGER VARCHAR2(30) NOT NULL,  
    LODGE_ADDRESS VARCHAR2(30));
```

```
CREATE TABLE SKILL_DETAILS(  
    SKILL VARCHAR2(30) PRIMARY KEY,  
    SKILL_DESC VARCHAR2(30));
```

```
CREATE TABLE EMP_DETAILS(  
    EMPNO NUMBER PRIMARY KEY,  
    FIRST_NAME VARCHAR2(30) NOT NULL,  
    LAST_NAME VARCHAR2(30) NOT NULL,  
    LODGE_NAME VARCHAR2(30),  
    CONSTRAINT FK_LODGE FOREIGN KEY(LODGE_NAME)  
    REFERENCES LODGE_DETAILS(LODGE_NAME));
```

```
CREATE TABLE EMP_SKILL(  
    EMPNO NUMBER,  
    SKILL VARCHAR2(30),  
    GRADE NUMBER(10),  
    CONSTRAINT PK_EMP_SKILL PRIMARY KEY(EMPNO,SKILL),  
    CONSTRAINT FK_EMP FOREIGN KEY(EMPNO) REFERENCES EMP_DETAILS(EMPNO),  
    CONSTRAINT FK_SKILL FOREIGN KEY(SKILL) REFERENCES SKILL_DETAILS(SKILL));
```

2. Insert Data

```
INSERT INTO LODGE_DETAILS VALUES('Jessy Lodge','Rajan','Rakesh Nagar');  
INSERT INTO LODGE_DETAILS VALUES('Rajesh Lodge','Rajani','Anna Nagar');  
INSERT INTO LODGE_DETAILS VALUES('Gooday Lodge','John','Sea Road');
```

```
INSERT INTO EMP_DETAILS VALUES(101,'James','Jackson','Rajesh Lodge');  
INSERT INTO EMP_DETAILS VALUES(102,'Kalpash','Raj','Gooday Lodge');  
INSERT INTO EMP_DETAILS VALUES(103,'Jasmine','Joy','Jessy Lodge');
```

```
INSERT INTO SKILL_DETAILS VALUES('Oracle SQL','Oracle Corporation');  
INSERT INTO SKILL_DETAILS VALUES('Dot Net','Microsoft');  
INSERT INTO SKILL_DETAILS VALUES('Sybase','SAP Corporation');
```

```
INSERT INTO EMP_SKILL VALUES(101,'Oracle SQL',9);
INSERT INTO EMP_SKILL VALUES(101,'Sybase',8);
INSERT INTO EMP_SKILL VALUES(102,'Dot Net',9);
INSERT INTO EMP_SKILL VALUES(103,'Oracle SQL',8);
```

3. Update Lodge

```
UPDATE EMP_DETAILS SET LODGE_NAME='Gooday Lodge' WHERE EMPNO=101;
```

4. Delete Employee 103

```
DELETE FROM EMP_SKILL WHERE EMPNO=103;
DELETE FROM EMP_DETAILS WHERE EMPNO=103;
```

5a

```
SELECT * FROM EMP_SKILL WHERE UPPER(SKILL)=UPPER('Sybase');
```

5b

```
SELECT E.FIRST_NAME,E.LAST_NAME,S.SKILL,E.LODGE_NAME
FROM EMP_DETAILS E JOIN EMP_SKILL S ON E.EMPNO=S.EMPNO;
```

5c

```
SELECT COUNT(*) AS NO_OF_SKILLS FROM EMP_SKILL WHERE EMPNO=101;
```

5d

```
SELECT E.FIRST_NAME,E.LAST_NAME
FROM EMP_DETAILS E JOIN EMP_SKILL S ON E.EMPNO=S.EMPNO
WHERE S.SKILL='Oracle SQL';
```

6. View

```
CREATE VIEW EMP_VIEW AS
SELECT E.EMPNO,E.FIRST_NAME,E.LAST_NAME,S.SKILL,E.LODGE_NAME
FROM EMP_DETAILS E JOIN EMP_SKILL S ON E.EMPNO=S.EMPNO;
```

7. Synonym

```
CREATE SYNONYM EMPV FOR EMP_VIEW;
```

8. Index

```
CREATE INDEX IDX_EMP_SKILL ON EMP_SKILL(SKILL);
```